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Real-Time Monitoring of Driver Distraction: State-of-the-art and Future Insights

Eva Michelaraki^{*1}, Christos Katrakazas¹, Susanne Kaiser², Tom Brijs³, George Yannis¹

¹National Technical University of Athens, Department of Transportation Planning and Engineering,
5 Heroon Polytechniou str., GR-15773, Athens, Greece

²KFV, Austrian Road Safety Board, Schleiergasse 18, 1100 Wien, Austria

³UHasselt, School for Transportation Sciences, Transportation Research Institute (IMOB),
Agoralaan, 3590 - Diepenbeek, Belgium

*Corresponding Author: evamich@mail.ntua.gr

Abstract

Driver distraction and inattention have been found to be major contributors to a large number of serious road crashes. It is evident that distraction reduces to a great extent driver perception levels as well as their decision making capability and the ability of drivers to control the vehicle. An effective way to mitigate the effects of distraction on crash probability, would be through monitoring the mental state of drivers or their driving behaviour and alerting them when they are in a distracted state. Towards that end, in recent years, several inexpensive and effective detection systems have been developed in order to cope with driver inattention. This study endeavours to critically review and assess the state-of-the-art systems and platforms measuring driver distraction or inattention. A thorough literature review was carried out in order to compare and contrast technologies that can be used to detect, monitor or measure driver's distraction or inattention. The systematic review followed the Preferred Reporting Items for Systematic Reviews and Meta-Analysis (PRISMA) guidelines. The results indicated that in most of the identified studies, driver distraction was measured with respect to its impact to driver behaviour. Real-time eye tracking systems, cardiac sensors on steering wheels, smartphone applications and cameras were found to be the most frequent devices to monitor and detect driver distraction. On the other hand, less frequent and effective approaches included electrodes, hand magnetic rings and glasses.

Keywords: Distraction, attention, state-of-the-art technology, inattention monitoring systems, driver state monitoring, PRISMA.

1 Introduction

Approximately, 1.3 million people die every year on roads worldwide, with millions more sustaining serious injuries and living with long-term adverse health consequences (WHO, 2018). Globally, road traffic crashes are one of the leading causes of death, especially among children and young people, as well as the number one cause of death among those aged 15-29 years. Currently, road crashes are estimated to be the 8th leading cause of death across all age groups, and are also predicted to become the 7th dominant cause of death by 2030 (WHO, 2018).

Several human factors have been identified which affect the likelihood of a road traffic crash or a serious injury, but among them, driver distraction or inattention are some of the major contributors to road traffic fatalities and injuries (Brookhuis and De Waard, 2010; Fairclough et al., 2005; Papantoniou et al., 2017). It, therefore, becomes evident that limiting the exposure to these risk indicators is essential and critical to reduce traffic injuries and promote road safety.

Specifically, driver distraction (either from in-vehicle or external stimuli) represents an important factor of driver state with negative impact on road safety and is a major cause of vehicle crashes worldwide (Regan et al., 2008). At the same time, technological developments make massive and detailed operator performance data easily available, via new in-vehicle sensors that capture detailed driving style (Oussous et al., 2018). This creates new opportunities for the detection and design of customized interventions to mitigate the risks, increase awareness and upgrade driver performance, constantly and dynamically (Horrey et al., 2012; Toledo et al., 2008). The optimal exploitation of these opportunities is the challenge that the i-DREAMS project faces.

Road safety is increasingly threatened by distracted driving (e.g. involving a driver not paying sufficient attention to the road in spite of the presence of obstacles or other people). Several studies have revealed that there is a significantly increased risk for a driver of being involved in a car crash due to cognitive distractions (e.g. the driver is not focused on the driving task), manual distractions (e.g. hands are off the wheel for other non-driving activities) or visual distractions (e.g. not watching the road). The purpose of this study is to identify and assess the most important information for the parameterization of driver behaviour and the development of technology-based interventions and systems. The relationship between driving distraction and road safety has fuelled a dialogue that includes not only medical or psychological, but also economic, political and social points of view. Thus, this article proposes a systematic framework based on a human-machine system approach to identify all of the components, technologies and interactions of driver distraction.

The overall objective of the European H2020 i-DREAMS¹ project is to define, develop, test and validate a context-aware safety envelope for driving in a 'Safety Tolerance Zone' (STZ), with a smart Driver, Vehicle & Environment Assessment and Monitoring System. Taking into account, on the one hand, driver background factors and real-time risk indicators, and on the other hand, driving task complexity indicators, a continuous real-time assessment will be created to monitor and determine if a driver is within acceptable boundaries of safe operation (i.e. within the STZ). Testing and validation will be applied to car, bus and truck drivers as well as to tram and train drivers.

Within the above framework, the aim of the work documented in this paper is to review and assess state-of-the-art in-vehicle approaches and technologies as well as the various driver recording tools to monitor driver distraction and inattention. To achieve this objective, a comprehensive literature search (scientific as well as grey literature) was conducted. Identified measurement methods and associated platforms were assessed based on pre-defined criteria such as intrusiveness and effectiveness among others. The review was conducted from a transportation mode perspective, beginning with car systems which were covered most extensively in literature.

¹Further general project information can be found on the website: <https://idreamsproject.eu>

The paper is structured as follows. In the beginning, the overall objective of the i-DREAMS project as well as the aim of this research is provided. Subsequently, the theoretical background of driver distraction definition and corresponding indicators is given. This is followed by a section, in which, several definitions with regards to the terms of driver distraction and inattention are presented and the types of driver distraction are also analyzed. Moreover, the methodological approach of the current research is presented. The main part of the paper is covered by an extended literature review was carried out regarding all available state-of-the-art technologies of assessing driver distraction. In the next step, the results of technologies and systems that have been identified for the real-time monitoring of driver inattention are presented. Finally, overall conclusions for the continuous monitoring of driver distraction are highlighted in order to assist researchers and practitioners.

2 Defining distraction and corresponding indicators

2.1 Definitions of driver distraction

The use of different, and sometimes inconsistent, definitions of driver distraction can confuse researchers and road safety stakeholders (Regan et al., 2007; Regan et al., 2008). Definitions currently utilized in the literature are outlined either from individual scientific studies, or official bodies and organisations.

To begin with, the International Standards Organisation (ISO) defines driver distraction as "attention given to a non-driving related activity, typically to the detriment of driving performance" (ISO, 2004). In addition, Young et al. (2007) considered that distraction is defined as "the presence of an event, activity, object or person within or outside the vehicle which compelled or tended to induce the driver's shifting attention away from the driving task". At the same time, Stevens and Minton (2001) proposed driver distraction as "physical events, actions or conditions, in or on the vehicle that divert attention from driving".

According to Streff (2000), distraction involves "a shift in attention away from stimuli critical to safe driving toward stimuli that are not related to safe driving" while Regan and Strayer (2014) claimed that distraction can be defined as "a diversion of attention away from activities critical for safe driving toward a competing activity".

As there is no consistent definition for driver distraction and driver inattention across studies, the comparison among them, may be difficult or sometimes impossible. Even seemingly, similar works sometimes investigate slightly different concepts or measure different outcomes. It is worth noting that inconsistent definitions may also lead to disparate results of road crash data and therefore, to contrasting estimates or assumptions of the role of distraction in road crashes. Consequently, these concerns highlight the need to develop a common, generally well-accepted definition of driver distraction.

For that purpose, driver distraction can be defined as "a diversion of attention from driving, because the driver is temporarily focusing on another event, task, object or person which is not related to driving" (Regan et al., 2008). As a result, the driver's awareness, decision-making ability as well as driving performance are reduced, leading to an increased risk of corrective actions, near-crashes or crashes. Following the definition above, the current study focuses on identifying the ways in which distraction and inattention can be monitored during trips and less attention is given to the relationship between driver distraction and road safety. For instance, Papantoniou et al. (2017) and Cunningham & Regan (2018) provided two very interesting approaches with a review of driving performance parameters critical for distracted driving with regards to road safety (Kaiser et al., 2020).

2.2 Thresholds

The simplest method for the detection of driving distraction is thresholding, in which a certain feature value is compared with a preset threshold. It should be noted that thresholds of driver's distraction, drowsiness or risk tolerance might change in different driving situations. For instance, drivers probably keep different risk tolerance levels in urban areas or in highways. Furthermore, their mental or health state might affect this parameter, as well. It is worth mentioning that they might accept higher risk levels when they are in a rush or they are furious or nervous in comparison with their normal tranquil moments.

It should be noted that a certain degree of distraction is normal in co-driving of human-machine system; thus, the threshold values of the indicators may be different from those in traditional manual driving. The human-machine system approach, which involves the use of technology such as sensors and cameras, has a lower threshold for detecting driver distraction compared to traditional manual driving, which relies on the driver to recognize the signs of distraction. This is because technology can detect subtle changes in driving behavior or eye movements that may not be perceptible to the driver. In addition, the human-machine system approach can provide continuous monitoring, whereas traditional manual driving relies on periodic self-assessment. This means that the system can detect distractions that occur over a longer period of time or those that are intermittent. Overall, the use of human-machine systems can provide more accurate and reliable detection of driver distraction, which can ultimately lead to improved safety on the road. However, there is limited evidence based on literature review with regards to exact thresholds for particular distraction types for both human-machine system and traditional manual driving.

With respect to the human-machine system approach, the threshold values can vary depending on the system being used and the context in which it is being applied, while in traditional manual driving, the threshold values for some indicators can vary depending on the situation (e.g. speed, driver's experience) or the specific driving context (e.g. road conditions, weather). Nevertheless, some commonly used thresholds for identifying driver distraction are described.

To begin with, Tabrizi et al. (2009) detected drowsiness by applying a constant threshold to the PERCLOS value. Furthermore, Chang et al. (2018) claimed that continuous person's eye closure is typically in a range of 0.2 to 0.4 seconds to blink for traditional driving. It was also demonstrated that it might take more time, exceeding 1 or 2 seconds, signaling a drowsiness situation. On the other hand, Xu et al. (2018) found a threshold of eye closure duration of 0.15 seconds for detecting driver distraction based on a human-machine system approach. Regarding the eyes off road, in a human-machine system, the technology is designed to detect distractions immediately and alert the driver. This can happen even if the driver's eyes are off the road for just a few seconds. For instance, Klauer et al. (2006) revealed that eyes off road totaling more than 2 seconds for any purpose increase crash risk or near-crash by at least two times that of normal, baseline driving. On the other side, in traditional driving, a driver's eyes may be off the road for a longer period of time before they become distracted or lose focus. Moreover, Kashevnik et al. (2020) indicated that glances occur 4 seconds in parking area (e.g. speed less than 20 km/h) and 2 seconds if speed is more than 80 km/h. Interestingly, it was revealed that the threshold for glances is typically around 2 seconds for traditional driving, while for human-machine systems, it is around 1 second.

It is worth highlighting that for hand position, there was no threshold information. Only Kashevnik et al. (2021) demonstrated that there is a 15-second rule for driver information systems which specifies the recommended maximum time for the driver to perform manual operations in the vehicle with visual displays. Although vehicle designers should make cases for lower task limits, these 15 seconds can be considered as an indicator of distracted driving if drivers are engaged in manual vehicle controls, such as navigation for longer than 15 seconds.

2.3 Types of driver distraction

In general, there are four distinct types of driver distraction: visual, auditory, biomechanical and cognitive (Breen, 2009). To begin with, visual distraction involves taking one's eyes off the road, such as reading a message on the mobile phone, looking at a GPS system, looking away from the roadway, or checking a child's seat belt in the rear view mirror (Dragutinovic and Twisk, 2005). In

addition, auditory distraction is caused by the amount of time that the driver is hearing something not related to the driving task, like getting startled by the initial ringing of the cell phone or by the conversation itself (Scott and Gray, 2008).

Moreover, biomechanical distraction occurs when a driver is doing a physical action that is not related to the driving task or is manipulating something other than the steering wheel, such as manually adjusting the radio volume, eating or drinking, reaching for something out of the driving position, searching for an item in the bag or grooming (Fernández et al., 2016). Lastly, cognitive distraction occurs when the driver is not mentally focused on the act of driving, as two or more mental tasks may be performed at the same time, such as taking one's mind off of driving through listening and conversing and being lost in thought (Angell et al., 2015). It should be noted that in reality, some of the above types of distraction may be combined, for example when texting and taking on the mobile phone.

There are several types of driver's distraction, such as cognitive distraction, texting or talking on the mobile phone using the right/left hand while driving, operating the radio, drinking, reacting to warnings, conversation with passengers, etc. Research on prevalence of distracting activities has indicated that car drivers spend about 25-30% of total driving time on distracting activities. More specifically, the use of mobile phone while driving consists one of the most common driver distractions (Oviedo-Trespalacios et al., 2016). Stutts et al. (2003) revealed that 30% of car drivers used a mobile phone while driving, while Overton et al. (2014) indicated that mobile phone distraction resulted in 18% of fatal crashes and 5% of injury crashes. Based on the European Road Safety Observatory (ERSO, 2015), talking with a passenger, listening to music (radio), eating or drinking frequently occur while driving. More than 90% of car drivers listens to music (radio) and a high percentage (47.5-95%) handle a music device while driving. Between 40 and 80% of drivers talk with a passenger and about 50 to 80% eat or drink inside the car. Moreover, about a third use phones hands-free and 15-35% of car drivers sends or reads text messages while driving.

2.4 Physiological indicators

As real-time measurement of physiological and behavioural indicators is crucial (especially for the i-DREAMS concept), the most important indicators will be introduced below with definitions and descriptions. In general, physiological measures correspond to the continuous measurement of the physical responses of the body, such as heart rate or heart rate variability. The most reliable and sensitive physiological measures include eye movements, such as eye blink rate, blink duration, fixations, saccades and interval of closure as well as head movements, such as rotation and orientation. Among the physiological indicators, the individual's level of arousal is a central concept which is linked to distraction and attention (Borghini et al., 2014).

2.5 Behavioural indicators

Moreover, driver behaviour patterns can be a result of the driver's state and measured in real time. Specifically, behavioural measures and real-time interventions are based on techniques of direct registration of driver ability to perform driving tasks at acceptable and safe levels, and properly maintain the vehicle on the road without colliding with other road users (Da Silva, 2014). For instance, lateral position deviation, longitudinal control, reaction time and gap acceptance are some of the most important indicators to assess driving performance.

As there are several driver monitoring systems that exist for monitoring driving performance, the selection of both physiological and behavioural measures for driver distraction and inattention research should be guided by the nature of the task examined. A range of driver distraction measures, as well as the indicators that have been used in recent studies to evaluate the impact of distraction on driving performance is provided in Table 1, including physiological (e.g. eye or head movements) as well as behavioural measurements (i.e. longitudinal control, lateral control, reaction time, gap acceptance).

Table 1. A range of driver distraction measures with their indicators

Driver distraction measures	Indicators
Physiological measurements	
Eye movements	Glances
(Jia and Tyler, 2019; Liang et al., 2007; Mabry et al., 2019)	Saccades
	Fixations
	Blinks
	Gaze direction
	Eyes-off-road-time
	Electrooculography (EOG)
	Percentage of eyelid closure time (PERCLOS)
	Percentage of time spent not looking ahead (PERLOOK)
Head movements	Rotation
(Hammoud et al., 2008; Huang et al., 2019; Ji and Yang, 2002; Morimoto et al., 2000)	Orientation
	Pose
Behavioural measurements	
Longitudinal control	Speed
(Manser and Hancock, 2007; Ranney et al., 2005)	Headway
Lateral control	Lateral position
(Brooks et al., 2005; Greenberg et al., 2003)	Steering wheel control
	Standard deviation of steering wheel angle
Reaction time	Perception response time (PRT)
(Caird et al., 2008; Hancock et al., 2003)	Brake response time (BRT)
	Time-to-collision (TTC)
Gap acceptance	Number of collisions
(Farah et al., 2007)	Gaps accepted

In order to monitor the driver state and detect driving distraction or inattention, several measurement indicators have been proposed, as shown in Table 1. For instance, visual features include eye blinking, eye closure, head pose or facial expression (Yin et al., 2009; Ji et al., 2004). On the other hand, non-visual features consist of brain activity, pulse rate and heart rate. Moreover, physiological signals (electrocardiogram (ECG), electromyogram (EMG), electro-oculogram (EOG) and electroencephalogram (EEG)) are used to detect driver inattention (Khushaba et al., 2010). Lastly, deviations from lane position, vehicle speed, steering movement, pressure on the acceleration pedal consist also important and reliable indicators to measure driving distraction (Liu et al., 2009).

2.6 Detection methods of driver distraction

To reduce vehicle crashes and improve road safety, a system that can classify driving distraction is highly desirable and has attracted much research interest in recent years. In fact, there is a variety of distracted driving recognition algorithms or systems, using machine learning, deep learning, cognitive models, etc. Various traditional machine learning techniques have been implemented in order to detect driver distraction. For instance, Begum et al. (2021) examined the detection of driver cognitive distraction in real-time (i.e. blink rate, eye closure and yawning) using deep learning concepts and machine learning classifiers which helps in classifying the image of drivers into the different classes and predict if the drivers are in distracted state or not. In addition, Jin et al. (2012) applied Support Vector Machine (SVM) using vehicle control data as feature vectors in order to detect cognitive distraction, while Craye & Karray (2015) proposed a Hidden Markov Model (HMM) based method to detect distracted driving activities (e.g. driver's face and right arm). Considering the inherited uncertainty associated with the face features, Bayesian networks were used by Ji et al. (2004) to determine the probability of driver distraction and model the uncertainty. Lastly, Eskandarian & Sayed (2005) detected driver drowsiness by combining facial data with steering wheel movements using Neural Networks (NNs).

One of the most important aspects of the distraction recognition systems is to ensure the accuracy and the effectiveness of the corresponding recognition platforms and distraction warnings. In particular, Convolutional Neural Networks (CNNs) have achieved top performance (up to 80%) in

the competition (Colbran et al., 2016) while Abouelnaga et al. (2017) proposed a CNN system which achieved a 95.98% driving posture classification accuracy. Similarly, Koesdwiady et al. (2017) proposed a ConvNet VGG-19 framework, which has achieved the highest test accuracy of 95% and an average accuracy of 80% per class. What's more, rule-based algorithms and Support Vector Machines (SVMs) were applied and satisfactory results were accomplished as 80% accuracy in detecting visual distraction and 68-86% accuracy in recognition of cognitive distraction was identified (Masood et al., 2020).

Several studies have been conducted in order to examine driving performance measures through driving simulators. Undoubtedly, driving simulators refer to a controlled, safe and relatively realistic driving environment (Papantoniou et al., 2017). Nevertheless, based on the fact that driving simulators vary substantially in their characteristics, this can affect the validity or the fidelity of the results obtained. To that end, many researches have carried out comparative test studies, comparing the data of real car experiment with that of driving simulation test in order to analyze the effectiveness of distraction or other driving behaviours. Engström et al. (2005) examined the validity of driving simulator and the results revealed a good correlation between simulated and real-time driving performance. In addition, Agrawal et al. (2022) analyzed the psychological and cognitive effect of driving environment complexity by implementing a network-level driving simulator environment with real-world travel information.

It should be noted that several hardware and software algorithms are being developed so as to monitor driver distraction and inattention using the aforementioned measurement indicators, but they are tested mostly in the simulated environments instead of real driving ones. This was due to the danger of testing drowsiness in real driving environments. However, this forms the motivation of the current research which aims to assess and review the state-of-the-art technologies and systems that can have been used and tested not only in a simulated environment but also in real driving conditions.

Taking into consideration the purpose of the current study, the identification of the most reliable technology that monitors driving behaviour and alert drivers when they are drowsy or in distraction state will benefit society, science and humanity. In addition, systems that are able to predict unsafe driving behaviours in advance would contribute to safe driving and prevent road crashes. A reliable and robust driver distraction detection system would send an alert to the driver and thus reduce the number of hazardous situations on the road.

It is observable that the majority of the literature on monitoring distraction and inattention focuses on both physiological (e.g. eye and head movements) as well as behavioural indications (e.g. longitudinal and lateral control, reaction times and gap acceptance) which is due to the fact that these indicators are easier to capture with state-of-the-art equipment.

All in all, driver distraction is a multidimensional phenomenon and there is not a unique driving performance measure which is able to capture all effects of distraction. In addition, visual distraction has been found to have a greater effect on lateral control measures, whereas cognitive distraction affects the visual scanning behaviour more. Among all trackable parameters, longitudinal and lateral control measurements, surrogate safety measures such as reaction time or gap acceptance, and eye or head measures are deemed to be the most crucial to identify driver distraction. However, the diversity in the measures used, in combination with the diversity in experimental design (i.e. with regards to the road and traffic factors examined), often complicates the synthesis of the results, especially for less commonly examined distraction factors.

3 Method

The current study endeavours to critically review and assess the state-of-the-art systems and platforms measuring driver distraction and inattention. Towards that aim, a thorough literature review was carried out in order to compare and contrast technologies that can be used to detect, monitor or measure driver's distraction or attention. Studies have to fulfil specific selection criteria in order to be included in this review. In particular, the criteria set demands with regards to recent

publication time, study quality and relevance, as well as the overall validity and trustworthiness of the publication source. Previous similar works relevant for setting criteria for meta-analysis were considered as well (Martensen et al., 2016; Papadimitriou et al., 2019).

3.1 Literature Search

In order to review and assess the state-of-the-art attention and distraction measurement techniques, a systematic search of relevant scientific and grey literature was carried out using the search terms documented in Table 2. Although there was a range of studies investigating the impact of attention and distraction in the context of road safety, this literature search and review explicitly focused on research relating to objectively measuring and detecting driver distraction and inattention during trips, preferably in real-time driving conditions.

The key terms were then entered into the databases, with the following inclusion criteria:

- Studies published from 2000 and onwards
- Studies including information on real-time monitoring systems of driver distraction in the title or abstract
- Search term included in title, abstract or key words
- Language: studies published as English
- Source: peer-reviewed journals before peer-reviewed conference papers before commercial reports/websites
- Importance: number of citations

The search was conducted in the databases ScienceDirect, ResearchGate, Scopus, PubMed and Google Scholar. Publications were duplicated, screened by title (917 publications) and then by abstract. Relevant literature was documented and summarized. The limitation was set to publications after 2000 and publications from peer-reviewed English language journals, conference papers as well as commercial reports, websites and articles were considered for inclusion. Additional key references were also added. Eventually, 29 publications were screened thoroughly and the literature predominantly concerned car driving.

Table 2. Terms used for systematic search of the literature

Key search phrase	Search terms	Screened papers	Included papers
Distraction	"distraction" OR "distracted" OR "inattention" AND "driver monitoring" OR "driver measure" OR "monitoring distraction" OR "real-time"	917	29

3.2 The PRISMA Procedure

This review was carried out during June 2022 and adhered to the Preferred Reporting Items for Systematic reviews and Meta-Analyses (PRISMA) steps (Moher et al., 2009) to identify relevant research articles and it was recently updated in order to cover more conceptual and practical advances in the methods of identifying, selecting, evaluating and synthesizing studies (Page et al. 2021). The initial search was performed in October 2021 and all databases were last accessed in June 2022.

To begin with, the search resulted in a total of 2,407 papers with 996 duplicates. Two reviewers independently screened all papers utilizing a general inclusion/exclusion assessment which was similar to the approach followed by Hawker et al. (2002). In case that older adults were not included in the studies examined, these papers were excluded. A total of 72 articles were selected for full-text review, including full papers, articles and commercial reports suggested from a subject matter expert. After the full paper review, an additional 43 articles and reports were excluded for not meeting the inclusion criteria. In total, 29 papers, commercial reports and websites were included in the final review. The PRISMA flowchart and the search strategy is depicted in Figure 1.

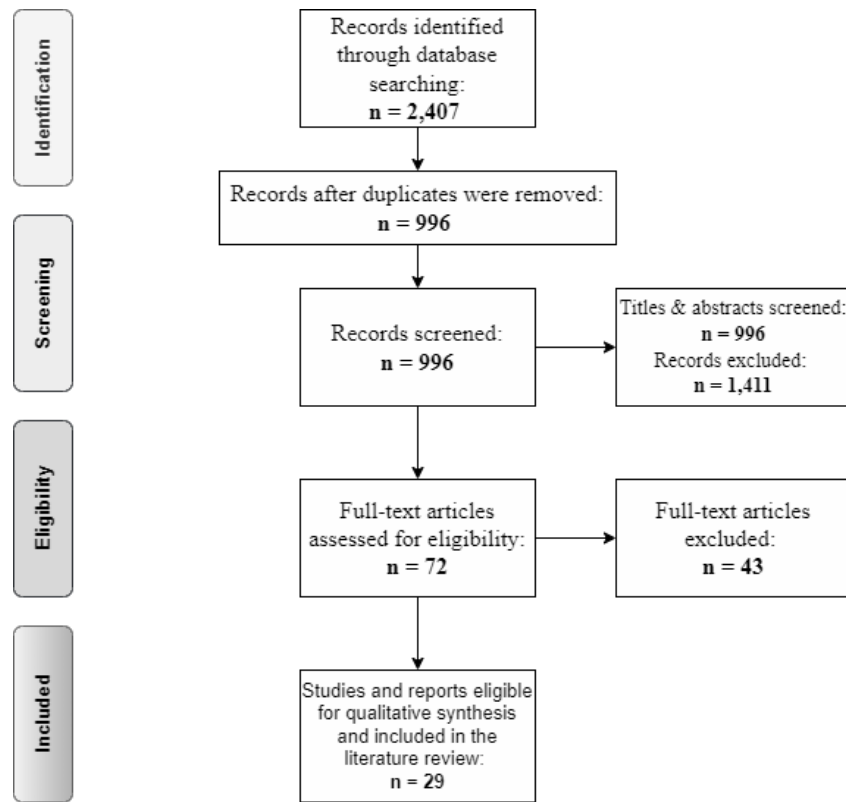


Figure 1. PRISMA flowchart of the systematic literature review

4 Results of the comprehensive literature review

The results of the literature review and commercial solutions revealed a variety of different sensors and systems that have been utilized to detect driver distraction. Particular emphasis of the review was given on commercially available products and technologies, which aim at promoting both road safety and eco-driving, as these remain the key interests of the i-DREAMS project.

The most prominent technologies that were applied, not only in the academic field, but also commercially applications, were found to be driver facing cameras (e.g. Delphi Electronics, Optalert, Mobileye), eye tracker systems (e.g. EyeAlert, SensoMotoric Instruments, Seeing Machines, Smart Eye, Vigo, Phasya) or glasses (e.g. Tobi eye-tracking glasses). In addition, particular emphasis was given to smartphone applications (e.g. CarSafe), wearable devices (e.g. BioRadio, BIOPAC, FlexComp, Shimmer 3, Empatica E4 Wristband) and steering angle sensors (e.g. Cardio Wheel, Texas Instruments Biometric Steering Wheel). The research literature documents two types of measures associated with periods of distraction or inattention: physiological and behavioural indicators.

4.1 Physiological indicators

In the past few years, many researchers have been working on the development of safety monitoring technologies using different techniques. To begin with, the “Driver Attention Monitor” by Toyota and Lexus has been conceived to detect driver attentiveness, using infrared sensors and cameras monitoring the driver’s face (Craye et al., 2016). This technology is able to identify the driver’s face orientation and facial expressions. With regards to the latter, previous works on detecting driver behaviour proved that facial movements provide useful information associated with secondary tasks, such as talking (Jain and Busso, 2011). In particular, features related to brow motion and eyelids movements can be used to capture signaling cognitive load (Li and Busso, 2013). Moreover, the system was found to be a non-intrusive solution for real-time distraction

monitoring, providing flashing lights and warning sounds. If no action is taken, the vehicle applies the brakes after sounding a warning alarm.

Fernández et al. (2016) and D' Orazio et al. (2007) proposed the EyeAlert system as an ideal technology which focuses entirely on the driver's alertness levels or distraction from the road ahead. When the infrared camera or sensors monitor the driver's eye closure rate, or blink duration and unsafe patterns are identified, an audible alarm is sounded. According to the available product information, the portable device focuses on the driver's inattention to the road ahead and it was revealed to be an effective technology which works regardless of weather or roadway conditions such as fog, snow, rain on and off highway. Furthermore, EyeAlert claims to sit easily on the dashboard and helps drivers make timely better choices and identify the duration and the length of their eyes closure.

Delphi Electronics, developed a real-time vision-based camera Driver Status Monitor (Edenborough et al., 2005). By detecting drivers' facial characteristics, this technology analyzed eye-closures and head pose in order to infer their distraction and inattention. In addition, the system found to be an effective and non-intrusive solution which provided real-time warnings and notifications and prevent drivers from being too distracted with non-driving tasks. Moreover, SensoMotoric Instruments (SMI) provided an eye-tracking system which measured gaze direction, head pose and orientation, PERCLOS, eyelid closure and blink, as well as pupil diameter and position (Juhola et al., 2013). It was revealed that SMI's InSight computer-based system was an effective way to monitor driver dis-traction, however user calibration was necessary.

Additionally, Seeing Machines is an effective and non-intrusive face and eye-tracking system which monitors the movements of a person's eyes, face, head, or facial expressions and distraction events in real-time through in-cab sensors and cameras (Zimasa et al., 2019; Wang et al., 2019). Similarly, Kuo et al. (2019) performed a naturalistic driving experiment in which participants drove a vehicle, which was equipped with the "Seeing Machines" driver monitoring system. It was clearly mentioned that these findings carried impact not only in terms of distraction and drowsiness but also in the context of naturalistic driving methodology, where real-time assessment of driver state can facilitate the analysis of large naturalistic datasets.

In the same way, Smart Eye is an eye-tracking system measuring eye fixation pattern, smooth pursuit of eye movement, blink rate and eyelid control through cameras on dashboard (Juhong et al., 2018; Brousseau et al., 2018). An exit survey was conducted by Kumar et al. (2016) revealed that Smart Eye device is a user friendly, cost-effective and easily accessible oculomotor monitoring tool and it did not appear to be an intrusive solution. As all of the participants claimed that they did not face any difficulty in under-standing the visual pursuit task, it was an effective technology for detecting driver inattention. What's more, the early warning distraction technology of Optalert, was found to be a non-intrusive solution, as drivers did not need to wear or do anything (Cori et al., 2021; Optalert, 2022). The product is designed to detect various signs, such as facial features, changes in the pattern of eye and head movements and skin conductance. A very interesting finding by Corbett (2009) revealed that drivers indicated that Optalert would be the preferred and effective option as a research tool related to distraction. Results also confirmed that this system leads to improved safety and warns the drivers when their distraction exceeds predetermined levels through visual and audio alerts. Moreover, Vigo is a Bluetooth headset, monitoring driver inattention at the wheel, by tracking eye and head motions (Laouz et al., 2020). Although it is an established product which can be implemented across fleets, it was revealed that the installation and training on use/analysis is necessary and minimal information is available. Drivers found this solution a bit intrusive as it required to wear their headset (Nahavandi et al., 2022).

Cardio Wheel, an Advanced Driver Assistance System found to be an effective and un-intrusive solution that acquired the electrocardiogram (ECG) from the driver's hands via sensors on the steering wheel to continuously detect distraction (Cardio Wheel, 2022). One of the most important ad-vantages of this technology is that it can be integrated with certain third-party systems, such as Mobileye and GeoTab, providing complete fleet management solutions for enhanced road safety (Cardio Wheel, 2022). Furthermore, Texas Instruments Biometric Steering Wheel is a non-intrusive

technology for measuring driver distraction but no information was found about the validity of this technology (Texas Instruments Biometric Steering Wheel, 2022). Texas Instruments proved a concept of how biometric sensors mounted on a steering wheel can be used to obtain important information from a driver in real-time, on condition that simple hand contact is required (Abu-Faraj et al., 2018). This product combines modern solid-state technology with low-power processing ability and wireless communication to detect respiration rate, pulse rate as well as ECG-based heart rate from a standalone system. According to the available technology information, it was found that it cannot be used in real-time conditions and it is not available for sale. However, it can only be available for testing in a simulator environment.

Phasya developed the first software portfolio related to the development, the test as well as the validation of several physiological and cognitive states monitoring systems influencing human performance (El Fassi et al., 2018). Phasya claims that software modules detect visual distraction based on the analysis of gaze direction, eyelids gap, pupil position, pupil diameter or eye images. By defining eye-lids movements and gaze areas of interest according to the environment and the use case, this technology enables to characterize the visual distraction in several applications. An interesting finding is that Phasya can offer a complete product for pilot trials and monitor several physiological and cognitive states of train drivers. As it enables to measure the distraction levels continuously, in real-time, automatically, it was found to be an effective solution, referring to validity and reliability.

BioRadio by Great Lakes Neuro Technologies is a wearable device that acquires physiological data for detecting driver distraction through electrodes attached to fingers (Chandrasiri et al., 2013). It is easy to set up and operate and the wearable wireless physiology monitor can stream data to a computer via Bluetooth or save it to memory for mobile monitoring. Results indicated that the wireless connectivity and the way it records signals was seamless and noise-free. In addition, BIOPAC System is also a wearable technology, using electrodes attached to the driver's body which can be placed once and then connected to the wearable recording system (Perry et al., 2011). It is easy to use, easy to wear, and provides a 24-hour data logger. Furthermore, FlexComp from Thought Technology is a wearable device using electrodes attached to fingers (Poh et al., 2009). Sensor locations can be set on non-intrusive locations on the body such as on forearm, as there are less intrusive than on fingers and less sensitive to movement. This technology is mainly used for bio feedback and the transferability to the driving context is not clear. As the above wearable devices are wireless, portable and easy to use technologies, they are a suitable solution for clinical research and teaching, but they can be utilized only with assistance of project staff in a simulator study and they are not appropriate for on-road tests.

In addition, Shimmer 3 provides a wearable device using wireless and robust body worn sensors (Burns et al., 2010). Specifically, photoplethysmography (PPG) ear clip using electrodes position on chest or arms, while a GSR unit, EMG3 using electrodes attached to two fingers (Shimmer 3, 2022). Shimmer 3 is a good solution for a simulator study as there are non-invasive and low-cost sensors, suitable for in car use. However, results have to be further validated with a larger sample as there is was only small sample size examined. Furthermore, Empatica E4 Wristband is a wearable device, equipped with sensors that offers real-time high-quality physiological data (Borrego et al., 2019). It was found to be an effective, easy to use and non-intrusive technology for the identification of driver distraction. The system's battery runs 48 hours and an internal memory allows to record for up to 60 hours of data. As a result, Empatica E4 is an ideal solution for longitudinal studies.

Hand sensors such as a hand magnetic rings or magnetic eyeglasses clips were found to be less frequent approaches in order to monitor driver distraction (Huang et al., 2019). Tobi eye-tracking glasses are less effective for monitoring driver distraction as the calibration of eye tracker might be time-consuming (Niehorster et al., 2020). Eye-tracking glasses are intrusive as drivers are required to wear them during driving. Results indicated that this technology was not suitable for on-road trials.

4.2 Behavioural indicators

Mobileye is a forward-facing camera system, which alerts drivers when an imminent rear-end collision is looming, helps to keep a safe following distance, warns then about unintentional lane departures, and provides indications about the detected speed limit signs. It was found to be an effective and non-intrusive solution for monitoring the adverse consequences of driver distraction, promoting road safety. Moreover, smartphones, with their embedded sensors, such as gyroscopes, accelerometers and magnetometers, were found to be promising tools for monitoring driving behaviour effects of distraction (Wahlström et al., 2017). Smartphone applications which can provide measures such as lateral and longitudinal acceleration, can be utilized for surrogate safety measures capturing observed distraction and inattention. For instance, You et al. (2013) presented a driver safety application, called CarSafe, which detects drivers to dangerous driving conditions as well as inattentive driving and alters the drivers accordingly. It uses a computer vision as well as machine learning algorithms on the smartphone to monitor and detect whether the driver was distracted using the front-facing cameras, while processing the data in real-time with the goal of minimizing missed events inside. However, this application is only available for Android phones.

It should be mentioned that since smartphones are portable devices, they are more related to the person who carries them, than to the car. This implies that these devices are not directly linked to the car structure, well-fitting many vehicle types. Smartphone solutions are increasing in vehicle telematics because they are scalable, upgradable and low cost (Papadimitriou et al., 2019). Also, they can provide instantaneous driver feedback and have many embedded sensors. Issues that have to be considered are the low quality of the sensors, which are not primarily selected for vehicular measurements. Moreover, smartphones are not fixed, leading to issues as regarding relative orientation, driver/passenger recognition and GNSS coverage.

To sum up, in order to monitor driver distraction and inattention, several hardware and software systems and technologies were examined. One of the main conclusions that can be drawn is that the most frequently utilized method for the continuous driver monitoring was found to be the use of physiological indicators. Eye movements such as the number and duration of eye fixations as well as ECG measures and head movements are indicated to be the most reliable ones. It is worth mentioning that the majority of the studies reviewed, were conducted and tested mostly in driving simulated environments with limited studies using open field driving experiments with real road conditions within a specific transport mode.

This result is plausible due to the danger of testing inattention on road driving environments, given the ethical constraints that come with inducing distraction. What's more, a manipulation check is easier to conduct in the controlled environment of a simulator. However, the results obtained in a driving simulator study may be applied to a real traffic environment. In addition, in driving simulators, there was not found a particular technology, device or navigation system which was directly connected into the vehicle for distraction monitoring. For instance, no product was able to discriminate between cars' or trains' interior. Consequently, all methods that were developed from driving simulator experiments in order to measure distraction and inattention were easily transferable to different transport modes. Table 3. Overview of devices and technical equipment used in the reviewed studies on measuring distraction and inattention provides an overview of devices and technical equipment used in the reviewed studies on measuring distraction and inattention.

Consequently, during the review of the existing and available technologies to monitor driver distraction, several and different kinds of sensors and systems have been identified. Eye-tracking systems such as Seeing Machines, SensoMotoric Instruments SensoMotoric, Toyota and Lexus' Driver Attention Monitor and Vigo were found to be effective and non-intrusive solutions for monitoring driver distraction and inattention. In addition, cameras on dashboards such as Optalert, Delphi Electronics Driver Status Monitor, EyeAlert and Smart eye detected and tracked lane boundaries, by connecting to the vehicle's CAN bus. Mobileye used a forward facing camera to detect the headway to the lead vehicle. As cameras can be installed in the majority of vehicles, they were non-invasive technologies with low cost and operation power. Nevertheless, limitations included poor visibility issues, such as shadows, bad lighting and weather. Moreover, sensors on steering wheels such as Cardio wheel and Texas Instruments Biometric Steering Wheel were the

most effective ways to detect driver inattention. In-vehicle OBD-II sensor systems has been tested both in city contexts and on highway segments and has been applied to both cars and light duty vehicles.

Furthermore, smartphone applications, such as CarSafe, were found to be portable, scalable, upgradable and low cost solutions to measure driver distraction. Electrodes attached to the body (i.e. finger, arm), such as BioRadio by Great Lakes NeuroTechnologies, BIOPAC Systems, BioNomadix® Logger, FlexComp from Thought Technology, Shimmer 3, including a PPG ear clip or Shimmer 3 galvanic skin response (GSR) unit, EMG3 were found to be portable and easy to use technologies, consisted a good solution for clinical research and teaching, but they are not suitable for real-time driving processing. It should be noted that these technologies are intrusive since they required electrodes to be attached to the drivers. Empatica E4 Wristband was revealed as an effective, easy to use and non-intrusive technology for the identification of driver distraction, while Phasya was found to be a software technology with no equipment used to measure aspects of driver state. Tobi eye-tracking glasses provided medium intrusiveness, not suitable for on-road trials and hence, assessed as a less effective solution for monitoring distraction. Lastly, hand sensors such as hand magnetic rings or magnetic eyeglasses clips were found to be less frequent approaches in order to monitor driver distraction.

Table 3. Overview of devices and technical equipment used in the reviewed studies on measuring distraction and inattention

Product/ Technology	Equipment, measurement method	Intrusiveness	Simulator (Yes/No)	On Road Test (Yes/No)	Indicators	Overall assessment and considerations for i-DREAMS project (+ advantages/ - disadvantages)
Physiological measurements						
Seeing machines (Zimasa et al., 2019; Kuo et al., 2019; Wang et al., 2019)	In-cab sensor, cameras facing forward and driver	Contact free	Yes	Yes	Face and eye tracking indicators	+ Used in truck/car fleets + Planned to be used on UK rail network + Already implemented in trams following tram crash + Can be designed to issue alerts + Established product + Combine multiple camera sensors to detect a wider range of movements - Would need installation and training on use/analysis - No clear results on time headway The research findings highlighted the capability of EEG delta and alpha spectrum toward characterizing microsleep events; therefore, with consideration to user acceptability, application toward drowsiness detection is plausible via EEG electrodes embedded in the typical headset.
Vigo (Laouz et al., 2020; Nahavandi et al., 2022)	Head, eyes tracking, headset	Medium	Yes	Yes	Vision based: 20+ eye /head tracking parameter	+ Real-time performance monitoring + Can be implemented across fleets + Established product - Would need installation and training on use/analysis - Requires driver to wear headset - Minimal information available It was revealed that the methods for measuring the driver behavior give high-resolution results in the case of good lighting but these results become less good in the light of night because it suffers from the detection of facial expressions.
Optalert (Cori et al., 2021; Optalert, 2022; Corbett, 2009)	Video cameras on dashboard, steering wheel	Contact free	Yes	Yes	Vision based: eye tracking, facial features, amplitude and velocity ratio of blinks	+ Issues early warnings + Driver does not need to wear or do anything + Established product - Would need installation and training on use/analysis - Licensable software The findings of the pilot study suggested that there was potential for eye blink measures to be utilized to assess fitness to drive in obstructive sleep apnea (OSA) patients. Moreover, OSA patients were drowsier than healthy controls according to some of the continuous real time eye blink drowsiness measures.

Cardio wheel (Lourengo et al., 2015; Cardio Wheel, 2022)	Sensors on steering wheel	Low	Yes	Yes	ECG, Heart Rate Variability	+ Dashboard for fleets of vehicles + Can be integrated with certain third-party systems + Non-intrusive - Requires contact of both hands to steering wheel - Requires custom steering wheel The CardioWheel solution monitors the ECG while driving, extracting relevant information related with the driver's identity and health state.
Smart eye (Juhong et al., 2018; Brousseau et al., 2018; Kumar et al., 2016)	Eye tracking cameras on dashboard	Contact free	Yes	Yes	Vision based: eye, face and head tracking	+ Developed for automotive industry + Established product + Non-intrusive + Driver does not need to wear or do anything - Would need installation and training on use/analysis - Sensitivity of the system The eye movement was used to control cursor that appear on monitor and eye blink used for entering command. The system was good enough to be used for people with disability.
Phasya (El Fassi et al., 2018)	Any equipment used to measure aspects of driver state	Dependent on equipment used	Yes	No	Vision based, eye tracking: PERCLOS, facial features, blink duration, pupil diameter, facial images	+ Range of modules available + Offers test/validation of physiological and cognitive state monitoring systems + Can be used within rail and automotive industry - Software - Unsure of validation This study demonstrated that the Phasya monitoring device was able to measure drowsiness under real driving conditions. The results emphasised the need for prevention campaigns to include early detection of risk factors, as well as optimising the care of drivers suffering from drowsiness.
Texas Instruments Biometric Steering Wheel (Abu-Faraj et al., 2018; Texas Instruments Biometric Steering Wheel, 2022)	Sensors on steering wheel, measuring pulse, respiration	Low	Yes	Yes	ECG heart rate, pulse rate, respiration rate	+ Fully assembled board developed for testing and distraction validation + Non-intrusive - Requires contact of both hands to steering wheel - Unsure of validation - Not available for sale The originality of this device resides in the automation of the necessary tasks that a driver has to do in case of such emergency with minimum effort. Thus, the developed device provides an easy-to-use heart attack detection that helps people with a history of chronic heart diseases to avoid crashes
BioRadio by Great Lakes NeuroTechnologies	Electrodes attached to finger (in study: placed on steering wheel)	Medium	Yes	No	ECG, Electromyography	+ Wireless connectivity + Seamless and noise-free recorded signals + Used for clinical research - Signals with accuracy of only 70%

(Chandrasiri et al., 2013)						- Use only with assistance of project team Testing results showed that the performance of all parts of the EEG system satisfies the design specifications.
BIOPAC Systems, BioNomadix® Logger (Perry et al., 2011)	Electrodes attached to the body	Medium	Yes	No	ECG, Electrodermal Activity, temperature	+ Easy to use technology + Easy to wear technology + 24-hour data logger + Electrodes are placed once, then connected to the wearable recording system + Can be used with other devices (i.e. eye tracking) - Participants were students, university personal Biopac System was a reliable method for measuring sympathetic nervous system activity with minimum measurement variability.
Empatica E4 Wristband (Borrego et al., 2019)	Wristband with sensors	Low	Yes	Yes	Electrodermal Activity sensor	+ Easy to use technology + Battery runs 48 hours + Internal memory with up to 60-hour recorded data + Additional equipment: 3-axis accelerometer to capture motion-based activity, event-mark button - No details on heart rate parameters provided The key findings showed that correlation between both systems were low to moderate for negative and positive-valenced images, and non-existent for neutral images. These results suggest that the E4 should be used with caution when assessing changes in emotionally-valenced images and should be avoided for neutral images.
FlexComp from Thought Technology (Poh et al., 2009)	Electrodes attached to fingers	Medium	Yes	No	Heart Rate Variability, Electrodermal Activity	+ Sensor locations can be set on non-intrusive locations on the body + High quality signals + Easy to use technology + Use fiber optic for real-time monitoring - Mainly used for biofeedback - Transferability to driving not clear - License restriction is enforced through limitations on the software's functionality Photoplethysmographic sensors were integrated into regular earphones to provide discreet and non-stigmatizing heart rate measurements. Preliminary results indicated that the proposed system had a high degree of accuracy at rest with an average error of 0.63%.
Shimmer 3, including PPG ear clip (Burns et al., 2010)	Electrodes positioned on chest or arms	Medium	Yes	No	ECG	+ Electrodes are placed on participant fixed for each trial + Can be used with other devices - Used in laboratory research The results indicated that an exciting potential for Shimmer 3 was the

						introduction and application of new technologies supporting older people living independently.
Shimmer 3 GSR unit, EMG3 (Shimmer 3, 2022)	Electrodes attached to two fingers	Medium	Yes	No	Electromyography	+ Electrodes are placed on participant each trial + Non-invasive sensors + Low-cost sensors + Suitable for in car use - Results have to further validated with a larger sample - Small sample size - Used in laboratory research - Cannot be used together with Shimmer 3, which measures ECG The Shimmer GSR+ sensor monitors skin conductivity between two reusable electrodes attached to two fingers of one hand caused by a stimulus the sweat glands become more active, increasing moisture on the skin and allowing the current to flow more readily by changing the balance of positive and negative ions in the secreted fluid (increasing skin conductance).
Tobi eye-tracking glasses (Niehorster et al., 2020)	Eye-tracking light glasses	Medium	Yes	No	Vision based: mean fixation time	+ Very detailed documentation of operationalization, design and procedure - Calibration of eye tracker might be time-consuming - Eye-tracking glasses medium-intrusive - Not suitable for on-road real-time trials This system recorded binocular gaze direction and provided pupil size, the 3D orientation of each eye ball in a coordinate system fixed to the headset, and gyroscope and accelerometer data indicating movement of the headset.
EyeAlert (Fernández et al., 2016; D' Orazio et al., 2007)	Eye tracking cameras on dashboard	Low	Yes	Yes	Vision based: eye, face and head tracking	+ Visual feedback and auditory/voice alarms + Works in all weather and road conditions + Small and portable device + Mounts easily on the dashboard - Sensitivity of the system The EyeAlert system focused entirely on the driver's alertness levels or inattention to the road ahead, regardless of the time of the day or the weather conditions.
Delphi Electronics Driver Status Monitor (Edenborough et al., 2005)	Vision-based single camera	Low	Yes	Yes	Eye-closures and head pose	+ Real-time warnings and notifications + Non-intrusive + Increases road safety + Offers the most direct indication of early distraction - Sensitivity of the system This information was used to warn the driver and modulate the actions of other safety systems. The purpose of this monitor was to increase road safety by preventing drivers from falling asleep or from being

						overly distracted, and improve the effectiveness of other safety systems.
SensoMotoric Instruments SensoMotoric (Juhola et al., 2013)	Eye tracking system, computer vision-based	Low	Yes	Yes	Gaze direction, head position and orientation, eyelid closure, PERCLOS	+ Quality product + Usability + Validation to obtain solutions that are truly focused on resolving an unmet need - User calibration is necessary Basic skills like movement on plain and rough surfaces in forward, reverse direction and turning capability were analyzed for easier comparison with other existing wheelchair setups on the bases of controlling mechanisms, compatibility, design models, and usability in diverse conditions. Results demonstrated that the system successfully operated with average response time of 3 s for eye and 3.4 s for voice control mode.
Toyota and Lexus' Driver Attention Monitor (Craye et al., 2016; Jain and Busso, 2011; Li and Busso, 2013)	Eye tracking cameras on dashboard, sensors	Low	Yes	Yes	Face orientation and facial expressions	+ Provides flashing lights and warning sounds + Prevents frontal collisions, unintended lane departures and night-time crashes + Intuitive features for enhancement driver's awareness of surroundings - System may in some cases not operate properly due to a variety of road/vehicle/weather conditions - People and obstacles that show in the monitor differ from the actual position and distance The complementary information extracted from all the modules allowed a reliable estimation of driver inattention. The experimental results showed that fatigue has been detected with 98.4 % accuracy and distraction with 90.5 %.
Behavioural measurements						
CarSafe application (You et al., 2013)	Smartphone application, front-facing cameras, embedded sensors	Contact free	Yes	Yes	Lateral and longitudinal acceleration	+ Portable, scalable, upgradable and cheap device + Provides instantaneous driver feedbacks - Application is only available for Android phones - Low quality of the sensors - Are not fixed, leading to issues as regarding relative orientation, driver recognition and Global navigation satellite system (GNSS) coverage Results from the CarSafe deployment were promising -- CarSafe can infer a common set of dangerous driving behaviors and road conditions with an overall precision and recall of 83% and 75%, respectively. CarSafe was the first dual-camera sensing app for smartphones and represented a new disruptive technology because it provided similar advanced safety features otherwise only found in expensive top-end

						cars.
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5 Limitations

A few limitations can be arguably found in the current literature with regards to the review and overall assessment of state-of-the-art real-time technologies for monitoring driver distraction. First of all, one limitation lies in the nature of the design of the works themselves. As mentioned above, the majority of the researches and systems investigated, were tested mostly in simulated environments instead of naturalistic driving, probably due to the danger of testing inattention in real driving conditions as well as due to the problems of vision systems working in outdoor environments (i.e. lighting changes, sudden movements). Driver's physiological reactions or movements, distraction or inattention may be different in a driver simulator from those in real conditions. Also, simulations are known to underrepresent on-road conditions, making them less representative and effective solutions. Furthermore, drivers may not face the driving process seriously, or sometimes an extra discomfort may be added to participants, usually caused by simulator sickness.

Moreover, it is worth mentioning that most studies and devices examined, were used in cars and can be presumed to have been selected for their customizability. No study concerning professional drivers in heavier vehicles was found, therefore, the findings may not be as useful to other transport modes. It should be also noted that eye tracking systems or devices operating with cameras may not function properly when the driver is wearing sunglasses. Furthermore, their functioning may be dubious in extremely bright or poor light conditions as well as may be ineffective for people with very dark skin and atypical facial shape. There was also a noticeable lack of studies that focus on the indirect effects of driver distraction and inattention. For instance, the vast majority of eye tracking systems and cameras monitored the driver without detecting the road environment, such as a pedestrians crossing the street, road layout, traffic conditions, time of the day or weather (i.e. fog, snow, rain). Finally, it is worth noting that some technologies seemed very easy to use and handle with, but non-professional drivers and customers may be not able to buy them, due a high cost.

6 Future research directions

Undoubtedly, future research should examine different state-of-the-art systems, products and technologies testing more in real-time on road conditions, as the field validation would increase the reliability of the findings. Taking into account that simulators may provide contradictory, inconsistent and conflicting results and produce invalid research outcomes, systems should be validated in real conditions. Lack of detail, general limitations and underlying biases have to be relied upon to reach a conclusion and researchers must resort to identify which technologies for monitoring driver distraction or inattention are suitable to real-time naturalistic driving experiment. Due to the fact that each monitoring method or technique described above is not the one and only standard in research, a thorough testing in real-time conditions for different transport modes is indispensable. As all studies were conducted with devices resembling car interiors, therefore it could be beneficial to examine technologies that are able to detect driver distraction in buses, trucks, and trains or trams.

Moreover, with the exception of eye and head movements, various factors can also affect the driver's attention, so future studies shall consider alternative indicators, such as light variations, occlusions, daylight as well as noise. A next step should be the correlation between the traffic context and the driver's hazard or risk perception and situation awareness. Vision-based technologies and cameras can be used, not only for providing information inside the car related to driver, but also for extracting useful details outside the car, such as external conditions of the road ahead, traffic, road hazards, intersections, or even position regarding other cars. It is worth noticing that as the greatest risk occurs during twilight and night, all eye tracking systems and devices should work well under all conditions including if the driver is wearing sunglasses.

In future studies, technologies for monitoring driver distraction in real-time should be tested with bigger sample sizes and for longer periods of time. In this way, fuzzy knowledge base will be easier to be generalized. For instance, it would be a good idea to examine vision systems, especially in drivers wearing glasses in order to solve the problems for daytime operation. A combination of sensors, such as a steering wheel and a lateral position sensor in addition to the visual information would be beneficial to achieve a correct detection of driver distraction. Taking into account that the majority of the distraction recognition systems have been designed for different drivers (i.e. professional drivers and non professional drivers), it is expected that distraction indicators and characteristics may be different. Thus, future research should investigate distraction indicators, systems and measurements for professional drivers.

Finally, as the majority of measurements concerning distraction are frequently intrusive, in-vehicle personalization (Yi et al., 2019) or the correlation of driving behaviour data with visual measurements through advanced statistical or machine learning approaches could become beneficial. Such advanced methodologies have been found to be advantageous in many road safety aspects e.g. activity recognition (Azar et al., 2019), real-time conflict prediction (Formosa et al., 2020), and their practical and more generalized application on driver distraction could bring new insights in tackling the problem for safe driving (Valeriano et al., 2018).

7 Conclusions

The objective of this study was to review and assess state-of-the-art technologies and systems to monitor driver distraction and inattention. In addition, a selection of driver inattention factors including measurement methods were summarized and driver distraction indicators were reviewed.

Technologies and equipment used in the reviewed studies measuring inattention and distraction were separately reviewed and assessed in terms of intrusiveness and effectiveness and overall applicability for the i-DREAMS project purposes. An assessment of available technology was provided, focusing on the theoretical suitability of single devices or technologies for measuring the driver state constructs in question and the applicability in two settings 'simulator' and 'on-road trial'. Intrusiveness was the main reason for a negative assessment of a device for the on-road setting as well as prioritization of positive assessed devices was made in terms of effectiveness.

Currently, there is no standard procedure for measuring a driver's distraction, although a plethora of methods, indicators and algorithms has been established. In most cases, driver distraction was measured in terms of its impact to driver attention and driver behaviour (Papantoniou et al., 2017). The results of the literature review showed that the most frequent forms of distraction were found to be visual and cognitive distraction. Non-intrusive technologies were strongly preferred for measuring inattention, and vision-based systems have appeared to be the most attractive one not only for drivers but also for researchers. In particular, attention monitoring systems, including real-time head, gaze and eye tracking systems, sensors on steering wheels, smartphone applications, wearables and dashboard cameras were found to be the most frequent devices to monitor and detect the driver's distraction, with head position, viewing and scanning patterns and PERCLOS being the most reliable indicators.

On the other hand, less frequent approaches included hand magnetic rings and glasses. Non-intrusive methods were strongly preferred for monitoring distraction, and vision-based systems, providing physiological indicators, have appeared to be attractive for drivers. Nevertheless, complementing a specific technology with an Electrodermal Activity (EDA) measuring device, such as a Wristband or a (thermal) camera facing the participant will be beneficial, as the complementary method may provide evidence for validity.

Regardless of the measurement methods and their quality, practical considerations for implementations in i-DREAMS should be noted. The vast majority of reviewed literature and information concerned car driving. It was revealed that most of methods, technological devices and systems mentioned above can be easily transferred to all transport modes and no indication was found that contradict the assumption that the identified methods can be transferred from private cars to the other i-DREAMS modes: trucks, buses, trains and trams.

Wearable devices, such as eye tracking glasses were found to work only with the assistance of project staff in a simulator study and they were not available for on-road testing. It should be clearly mentioned that the impact on naturalistic driving conditions has to be considered when asking the participants to wear a device whenever they drive. For instance, when using cameras facing the participant, General Data Protection Regulation (GDPR) is to be considered carefully. Hence, with the exception of wearables, it can be concluded that attention monitoring systems are easily transferrable to all four modes of i-DREAMS. This could be very important for the project, providing flexibility, meaning that the system does not need to be redesigned for each mode of transport.

Systems aimed at increasing driver safety and prevent crash through real-time monitoring were examined. This issue will constitute the project's first pillar (i.e. real-time risk monitoring). Moreover, impact on driver safety can be expected to be higher, if proposed technologies in some way combine the local perspective (i.e. in-vehicle assistance with instant impact on driving) with the general perspective (i.e. longer-term support for a gradual change process in the vehicle operator). The development, implementation and testing of the most suitable technological solution (i.e. the i-DREAMS platform) could bring together these functionalities.

Within the i-DREAMS framework, the conclusions drawn from this study serve as the base for selecting appropriate measuring systems and devices for the future project work and for building the theoretical and mathematical model which are the backbone of the development of the i-DREAMS platform. Constructs to be measured are the driver's cognitive and affectional state (mental state) in terms of attention and distraction as well as more stable characteristics which are known to impact safe driving. Another outcome will be a research database with rich information of simulator and on-road drives of hundreds of participants. Since the database aims to facilitate future research, it can be argued that if a vast amount of data on test subjects is obtained, research findings will be better validated. However, this is a question of time, resources and also reasonableness towards participants volunteering to support the i-DREAMS research.

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